

COURSE : EL469 VLSI TECHNOLOGY

1. Course Title	VLSI Technology
2. Credit Structure	Lecture hours per week : 3 Tutorial per week: 1 Practical hours per week: 0 Total Credits: 3 Total Lectures 40-44 (each lecture @ 1 hr)
3. Course Code	EL469
4. Program/Semester	BTech/SEM VI and VII /PG
5. Category	Open Elective
6. Prerequisite courses	Nil
7. Instructor	Harpreet Singh Jattana
7. Foundation for	Advanced course in Nanofabrication
8. Abstract Content	The course is designed as per Industry requirements, covers details of various processes, techniques and methods employed in manufacturing of semiconductors, based on silicon or compound semiconductors like GaN, GaAs, AlGaIn and MEMS devices. The course deals with manufacturing process cycle for semiconductors. The focus is on knowing the Global semiconductor Industry and its impact on society. The details of various microfabrication techniques are covered from substrate preparation to wafer fabrication About substrate material - knowing the material property, Silicon as a material, Wafer Growth techniques like CZ, FZ etc, Clean room concepts, wafer cleaning techniques. Various fabrication techniques : Diffusion process and kinetics of diffusion, CVD and PVD, Deposition techniques and Furnace used, Ion Implantation techniques and equipment, Dry and wet etching, etch profile etc, wafer bonding.

	About wafer sort, device packaging, different types of packages, Bonding diagram. Device testing QA , failure mode and device reliability in brief.
Suggested Text book(s)	Book - VLSI Technology by SZ Lee Microfabrication - Plummer Industry Manuals Study materials will be provided

Detailed Course Contents		
Topic Name	Content (2 -3 lines per 4 – 6 lectures)	No. of lectures
Introduction to Semiconductor Industry, types of semiconductors and usage, markets, products	- Intro to Industry, market size, key companies, products, markets. Global scenario, Indian Scenario -Arena of Semiconductors -Elementary semiconductor, compound semiconductor intro (GaN, SiC, etc)	5
Material Properties, Silicon and wafer Growth	-Initial simple process flow for BiPolar, -Process flow for MOSFET -Crystal structures, properties, -Silicon properties -Wafer growth -Cz, Float zone -Types of wafers used in Industry.	5
Wafer specification, clean room concepts, wafer cleaning techniques in fabrication	-Specifications of wafers. -Key wafer metrology tools. -Clean room concepts, -Clean room classifications, =clean room, protocol, -wafer cleaning, -defects, dislocation. -Bearing on yield,	5
Deposition techniques	-Intro to deposition techniques -Thin films of nitride, oxide -Oxidation Mechanism -Growth kinetics, -Deal Groove model	5

	-About Furnace and Vacuum pumps -Dry and wet oxidation -Actual recipe of oxidation used in Industry -	
Photolithography Techniques, Light sources, types of litho, masks and reticle	-Photolithography: -Basics of OPTICS -The Litho process -Masks, Photoresists -Pattern Transfer -GDSII -Reticle design and mask-set -Optical Litho -Manufacturing methods and Equipments -Mask aligner	5
Stepper and Scanner NextGen Litho Techniques E-beam litho X-ray Litho DUV EUV	-g line -h line -DUV Litho -Next Gen Litho -OPC -RET -About Stepper and scanner used in low geometry process	5
Diffusion – Introduction, Process of diffusion Kinetics of Diffusion Ficks laws Well formation S/D Formation	-Diffusion: -Diffusion process -Models of Diffusion in solids -Ficks laws -Diffusion profile -Diffused wells	5

Ion Implantation The process Types of Implantors Models of implantation Deep implants	- Introduction to ion Implantation - -Process of Implantation - The Equipment basic details - - Models of process of Implantation - - Well implants - How to adjust V_t	3
Etching	-Etching: -Types of Etching -Isotropic and anisotropic -Wet etching of Si, SiO₂, Si₃N₄, Al -Etch stop -DRY etching: -Plasma assisted etching -Types of Etchants -Edge profile -Step Coverage -Etch mechanism	3
PVD	=Metallization -Integrated process flow at 130nm -Setting up V_t -Concept of HKMG	3

Outcomes and Objectives	1. spark an interest in Semiconductors / VLSI 2. To understand complete manufacturing cycle of Semiconductors 3. To Understand Product development cycle – specifications to qualified product 4. To learn techniques of Microfabrication 5. To learn methods of Silicon Ingot making 6. To understand mechanics of Oxidation and models representing Oxidation 7. To learn concepts of Lithography and Mask 8. To learn thin film Deposition techniques 9. To learn device package types and process of packaging At the end of the course, students will be able to understand the various techniques of Microfabrication used in manufacturing of Integrated circuits and MEMS devices
Grading Policy	Mid term test 40 marks End term examination , 60 Marks Total – 100 Marks Grades will be allotted based on GAUSSIAN DISTRIBUTION and not on absolute total marks.
 WIN_20231226_09_05_01_Pro.mp4 VIDEO ABOUT COURSE :	
Explanatory Notes	
1. Enter a discipline (see master list for guidance) and a first digit followed by xx: 1-4 BTech by year, 5 Mtech, 6 MSc, 7 Mdes	
2. Please indicate the program and desired semester in which it is to be offered. In 12, comment on what additional courses in that semester it should not conflict with.	
elective (BTech), elective (other programs)	
3. Appropriate categories: Core (all programs), technical elective, science elective, open	
4. Indicate the subsequent courses for which the course would be needed.	

5. The Instructor is permitted a certain degree of flexibility in each course. The following ratios of primary to optional content in terms of lecture hours are given as guidelines: Foundation Courses – 75:25, Track Core (Group Elective)/M.Tech. Specialization Core – 66:33, Electives – 50:50. For example, a 3-0-0-3 course is expected to have 40-42 lectures in the semester, and the primary content should specify 30 lecture hours worth of material in a Foundation Level course and only 20 hours of material in an Elective Course. The primary content is what the instructor must cover in the course; the remaining is at the discretion of the Instructor.
6. The outcomes and objectives should give on the one hand the skills and competencies acquired by the student as well as the value of the course towards industry/research/academic objectives.

POs-COs Matrix:

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
√	√	√	√	√	√	√					